

EV Charging Demand Forecasting Project Report

Introduction

The rapid growth of electric vehicles (EVs) has increased the need for efficient charging infrastructure and smart energy management systems. EV charging demand forecasting plays an important role in reducing energy wastage, preventing grid overload, and improving the overall utilization of charging stations. Accurate forecasting allows operators and energy providers to plan charging schedules, manage peak-hour demand, and optimize renewable energy usage. In this project, historical EV charging data was analyzed using data science and time-series forecasting techniques to identify patterns and predict future charging demand.

Abstract

This project focuses on predicting EV charging demand using historical charging behavior, time-related variables, and energy consumption patterns. Data preprocessing techniques such as missing value handling, datetime conversion, feature extraction, and rolling average generation were performed to prepare the dataset for analysis. Exploratory Data Analysis (EDA) was conducted using line plots and correlation heatmaps to identify trends and relationships between variables. Forecasting models including ARIMA and Prophet were implemented to estimate future charging demand. Model performance was evaluated using statistical metrics such as MAE and RMSE. The project demonstrates how predictive analytics can support efficient EV infrastructure planning and sustainable transportation systems.

Tools Used

- Python
- Pandas & NumPy
- Matplotlib & Seaborn
- Scikit-learn
- ARIMA Model
- Prophet Model
- Google Colab / Jupyter Notebook

Steps Involved in Building the Project

1. Imported the EV charging dataset and inspected the data structure.
2. Converted timestamp columns into datetime format for time-series analysis.
3. Handled missing values and arranged data chronologically.
4. Created time-based features such as hour, day, month, and weekend indicators.
5. Generated lag variables and rolling mean features for demand prediction.
6. Performed exploratory data analysis using line charts and correlation heatmaps.
7. Split the dataset into training and testing datasets.
8. Developed ARIMA and Prophet forecasting models.
9. Evaluated forecasting accuracy using MAE and RMSE metrics.
10. Exported forecasted results and prepared visual dashboards for interpretation.

Conclusion

The project successfully demonstrated the use of time-series forecasting techniques for predicting EV charging demand. The ARIMA and Prophet models effectively identified charging patterns and future demand behavior based on historical data. Data visualization techniques such as demand-over-time plots and correlation heatmaps provided meaningful insights into energy consumption and charging trends. The forecasting results can help charging station operators improve resource allocation, reduce operational costs, and support efficient grid management. Overall, the project highlights the growing importance of predictive analytics and intelligent forecasting systems in the development of sustainable electric mobility solutions.